Brandon Knutson

PhD Candidate — ML & Applied Mathematics

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EDUCATION

Colorado School of Mines — **Golden, CO**
PhD Computational & Applied Mathematics — *2022–Present*
Advisors: Dr. Daniel McKenzie & Dr. Samy Wu Fung

University of Colorado Boulder — **Boulder, CO**
BA Mathematics & Physics (with Distinction) — *2014–2018*

EXPERIENCE

Oak Ridge National Laboratory — **Oak Ridge, TN**
Machine Learning Intern — *2023*

- Trained XGBoost model on 898k buildings to predict heights from satellite imagery with 0.70m MAE, a 39% error reduction over baseline. Built geospatial data pipeline (PostGIS, QGIS, Docker); ablation study showed a new global height dataset added no predictive value.

Colorado School of Mines — **Golden, CO**
Instructor & Teaching Assistant — *2022–Present*

- Instructor of Record for Differential Equations for 60 students (2024) with course design responsibilities, after leading recitations and office hours as TA (2022–2023). Led a coding bootcamp for PhD students.

US Navy — **Norfolk, VA**
Operations Research Analyst (GS-11) — *2019–2022*

- Developed data-intensive cost models to evaluate acquisition proposals up to $4.1B. Briefed senior decision-makers across the Navy, Air Force, and NASA, securing hundreds of millions in savings. Secret clearance.

RESEARCH

- **Faster Implicit Network Training via Random Truncation** — *Manuscript in preparation*
  Theoretical and empirical analysis of a new class of training methods for implicit networks. Random truncation reduces computational cost while preserving convergence.
- **Logical Extrapolation in Neural Networks** — *AAAI 2026 (Oral)*
  First-author paper at AAAI, a top AI conference, benchmarking how implicit and recurrent neural networks generalize to harder, out-of-distribution mazes. Selected for oral presentation (top 4% of 23,680 submissions). Extra test-time compute yielded near-perfect accuracy on mazes 10× larger than training. Error analysis exposed shortcut learning that data diversification only partially fixed. *On Logical Extrapolation for Mazes with Recurrent and Implicit Networks*.
- **Open-Source Maze Generation Package** — *Journal of Open Source Software*
  Co-authored peer-reviewed Python package with 80+ GitHub stars, used by research community to generate datasets for studying neural network reasoning. *maze-dataset: Maze Generation with Algorithmic Variety and Representational Flexibility*.

SKILLS

Programming & Tools: Python, SQL, PyTorch, Pandas, Scikit-learn, MATLAB, Mathematica, LaTeX, Jupyter Notebooks, Git, Docker, AWS, Google Cloud, Linux, HPC, Multi-GPU Systems, AI Coding Assistants (Claude Code, GitHub Copilot)

Core Competencies: Deep Learning, Implicit & Recurrent Neural Networks, Bi-level Optimization, Stochastic Methods, Linear Algebra, Topological Data Analysis, Data Visualization

AWARDS & HONORS

2014–2018: GPA: 3.99, Dean's List, Phi Beta Kappa, Pi Mu Epsilon

2018: Marlene Massaro Pratto and David Pratto Scholarship (Mathematics)